CS 830 Project Proposal & results March 11th - Mar 31st 2025, May 8th 2025

CS 830 Introduction to Artificial Intelligence, Prof. Wheeler Ruml

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Modified March 23rd 2025 - March 31st 2025 based on review comments.
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Updated as per review comments and zoom video call.

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Goal:

"Can Large Language models (LLMs) say I don't know" or "inconclusive" given knowledge base by using resolution refutation, given its word sense disambiguation accuracy and world knowledge? LLMs

Why do we want to test the reasoning for inconclusiveness for Explicit reasoning tasks?

Resolution is refutation complete, if a set of clauses, that are in first order clausal normal forms, are unsatisfiable, then the resolution derives empty clauses and verifies unsatisfiability. But Resolution is NOT refutation complete if the set of clauses may be satisfiable, that is there is a clause that is a logical consequence of the set of clauses, as it may never terminate. If C is a first order clausal normal form that is unsatisfiable, then $C \vdash \perp$, since resolution combined with skolemization and classification is a refutation complete calculus for First Order Logic.[11] By Lifting lemma and Herbrand's theorem, proof for First-order resolution's refutation completeness can be proved. If a formula is satisfiable, proof search can terminate or go one forever, as there are infinitely many clauses that can be generated by resolution. Therefore resolution is not a decision procedure for First order logic. There may be other forms of resolution [12] but we only use conventional skolemization decomposition for this project. The conventional skolemization decomposition (CSD) are not meaning-preserving clausal forms. [13]

Large Language Models (LLMs) have shown remarkable reasoning and semantic understanding by effectively performing on tasks such as Natural Language Understanding [23]. Large Language models have been trained on various types of languages such as Natural languages, and formal languages such as Java, C, and Python [19]. More recent GPT-4o [18] can effectively process the textual, visual, and audio inputs. The Large Language Models that are trained on large amounts of data that learn distributional semantics fail on the basic commonsense reasoning tasks requiring analogy and logic [23]Large language models also do well in Multimodal AI tasks such as Visual Question Answering (VQA), Vision Language Retrieval (VLR), and Natural Language for Visual Reasoning (NLVR) than the other Multimodal AI systems [21]. Aside from the

computational costs and the api costs of applying LLMs to domain-specific tasks, the explicit-reasoning tasks still require hand-crafted domain-specific rules and logic that are difficult to scale and generalize to other applications [23]. The work [24] explores reasoning about how LLMs obtain answers and how they have difficulty during deductive reasoning. LLMs on the other hand, terminate and don't get stuck whether to conclude true, false or inconclusive, despite various other issues surrounding accuracy and planning during deductive reasoning. [7][15]

Large Language models (LLMs) serve as knowledge bases with world knowledge. [9] Large Language models do well in word sense disambiguation. [8] LLMs such as ChatGPT, GPTx models are able to translate natural language to first order logic and skolemized clausal normal forms.[4][7] LLMs are able to do deductive reasoning but don't do well in planning [7][15]. LLMs are known to have emergent abilities but recent works show evidence that emergent abilities are not fully emergent but are a combination of in-context learning, model memory and linguistic knowledge. [16]. More recent works on canonical versus non-canonical tokenizations demonstrate exponential growth of number of tokenizations as a function of sentence length.[17] The work on tokenization in LLMs [17] explores the fact that the LLMs are shipped with tokenizers that may deterministically encode text in canonical tokenization that are assigned with probabilities. The tokenization is not unique in LLMs i.e. the same word [Bat] can be tokenized as [B,at], [B,a,t], [Ba,t], [Bat]. The probability of [Bat] is the probability of its canonical sequence of tokens. Such a behavior is inherent to autoregressive LLM. [17]. The merging of tokens happens as follows: first word is split into characters and priority rules are applied. For example if each pair of characters in [Bat] generated, then a pair with maximum frequency of occurring together is chosen as a merge rule. This process seems to be repeated until desired vocabulary size is achieved, which is the Byte pair encoding (BPE). [17]. Merge rules are also dropped during training, while the rest of BPE is preserved. [17]. For each string x, there is a canonical tokenizer v* then canonical probability is p(x,v*). Determining by computing most likely canonical tokenization is hard as distributions of tokenizations of the word tokens shows that canonical tokens have higher marginal probabilities. This seems to emphasize the exponential increase in search space with that of canonical tokenization. The inference time of an LLM is when an input text is provided to an LLM for which the time LLM takes to tokenize and generate vector embeddings and process to generate an output. Thus [17] argues choice of canonical versus non-canonical can significantly guide the responses from an LLM and we can tap into conditioning such tokenization to guide LLM responses based on most-likely tokenization or marginal string probability. Two approaches that are explored in [17] show adjusting the marginal estimate over tokenizations that are not canonical and by fine tuning

hyperparameter α that adjusts weights for canonical versus non-canonical estimates. This work shows canonical tokenization ([B,at], [B,a,t]...) is the most likely tokenization which is also very close to the marginal probability of the text summed over tokenizations itself ([Bat]).

Problem statement

For this project, I attempt to test reasoning and results with a knowledge base for using Word Sense Disambiguation i.e. detecting correct meaning of the word, when it has multiple possible meanings (polysemy). The problem I attempt to test is "can LLM say I don't know" or "inconclusive" given knowledge base by using resolution refutation, given its word sense disambiguation accuracy and world knowledge? Given a knowledge base in natural language, generate clausal forms using LLMs and evaluate the clausal forms by resolution refutation algorithm and by also testing the clausal forms as inputs to LLMs and verify the results and compare.

I created two example knowledge base in natural language that can have multiple First Order Logic (FOL) interpretations. I generated clausal forms from Large Language Models in this case I used Claude i.e. claude-3.5-haiku20241022 & claude-2.7-sonnet-20250219 versions respectively and chatGPT. (An FOL interpretation is an assignment of meaning to the symbols of a formal language.). This is unlike assignment 6 and 7, since clausal normal forms (CNFs) are generated by Large Language models (LLMs) given a natural language knowledge base and those clausal forms serve as inputs to resolution refutation as well as they serve as input prompts back to the LLMs.

Word Sense Disambiguation:

Example1: Bat is a mammal. Bat is used in baseball.
Example2: Bark is a boat. Bark is a sheet of chocolate.
Domain of discourse: All words that have two different meanings {bat1 , bat2, bark1, bark2} where the meaning of each word is different. bat1 != bat2, bat1 != bark1, bat1!=bark2 and so on.

Prompt design follows Chain-of-thought prompt design used in [7].

Q: Each cat is a carnivore. Every carnivore is not herbivorous. Carnivores are mammals. All mammals are warm-blooded. Mammals are vertebrates. Every vertebrate is an animal. Animals are multicellular. Fae is a cat. True or false: Fae is not herbivorous. A: Fae is a cat. Cats are

carnivores. Fae is a carnivore. Every carnivore is not herbivorous. Fae is not herbivorous. True (Saparov A, He H 2023.)

Input prompts with knowledge base:

Prompt 1	Prompt 2	Prompt 3	Prompt 4
This is the knowledge base: Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bat flies with wings. Wings are not barrels. Barrels are not wings. Bats can fly. **There is a brown bat.** using clausal forms prove by resolution refutation if this statement is true or not or inconclusive	This is the knowledge base: Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bat flies with wings. Wings are not barrels. Barrels are not wings. Bats can fly. **There is a brown bat.** "Can we say Brown Bat can fly?" Is this statement is true or not or inconclusive?	Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bats have wings. Bats can fly. Flying requires wings. Wings are not barrels. Barrels are not wings. Bat can be elegant. There is a brown bat. **Elegant Brown Bat has a barrel and can fly.** using clausal normal forms prove by resolution refutation if this statement is true or not or inconclusive	Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bats have wings. Bats can fly. Flying requires wings. Wings are not barrels. Barrels are not wings. Bat can be elegant. There is a brown bat. **Elegant Brown Bat has a barrel and can fly.** Is this statement is true or not or inconclusive?

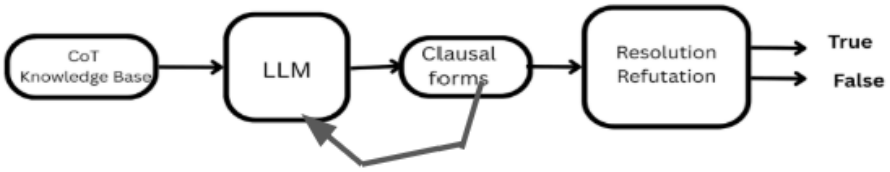
Method summary:

Input will be processed and converted to Skolemized CNF, unify over CNF and apply resolution refutation. (Unify will be part of resolution refutation).

Detailed Method Summary:

1) Input: knowledge base in natural language (as it was asked in the assignment 6 and 7)
2) Create a natural language knowledge base that can have multiple FOL representations, that is it can have multiple interpretations i.e. atleast 2.
3) Convert knowledge base to First Order Logic (FOL) representation 1 and 2 respectively (also without equality) (**using an existing work that I implemented a while ago**),
4) Generate Clausal normal forms from LLMs (claude 3.5, claude 3.7 and ChatGPT)
5) Apply Resolution refutation

6) Verify answers and compare with Classifical resolution refutation implementation



Results:

For this project, I manually generated a knowledge base with word sense disambiguation over a single word which is available in the prompt table in the following page as well as with full derivations in the Example Knowledge Base section. I tested this among claude-3.5-haiku20241022, claude-2.7-sonnet-20250219. Claude 3.5 and GPT.

- Resolution refutation sometimes sterminates but sometimes does not.
- I tested reasoning for true or false or inconclusive with resolution refutation and one without resolution refutation.

	Claude 3.5	ChatGPT	Claude 3.7	Resolution Refutation	**Correct answers**
Prompt 1 With resolution refutation	**True**	**True**	**Shows RR proof and also concludes Inconclusive with reasonable explanation**	**Does not terminate**	**True**
Prompt 2 Without resolution refutation	True	**Inconclusive**	**Inconclusive with reasonable explanation**	Does not terminate	**Inconclusive with word sense disambiguation**

Prompt 3 With resolution refutation	Inconclusive	Inconclusive	**False**	Does not terminate	**False, does not derive bottom**
Prompt 4 Without resolution refutation	True	Inconclusive	True	Does not terminate	**False with reasoning using word sense disambiguation**

Output (same as output format from assignment 7):

Verify if LLMs are able to emit results of inconclusive where applicable by appealing to word sense disambiguation as well as world knowledge. Compare results between LLMs, Resolution Refutation implementation with that of correct answers.

For prompts that said with resolution refutation, we want LLMs to not preserve meanings as well as justify why it could be inconclusive due to word sense disambiguation.
For prompts without resolution refutation, LLMs conclude inconclusive wherever applicable.

ClaudeAI and ChatGPT

ClaudeAI [6] and ChatGPT[5] are two large language models. ClaudeAI is built on ConstitutionalAI with self-reflection and feedback loop to learn from its responses to avoid generating harmful texts. ClaudeAI follows Reinforcement Learning with AI feedback (RLAIF) [6] while ChatGPT[4] follows Reinforcement Learning with Human Feedback (RLHF) [5]. Prompts are inputs to Large Language Models (LLMs) and they can be adjusted to get required output in specific formats. The LLMs are sensitive to prompts and tend to learn in-context [5]. Therefore applying prompt specific techniques to compare to that of traditional resolution refutation results would be easier.

Intuition and Motivation behind this problem (based on review comments):

GPTx models have been vastly tested for First order logic forms and translation from natural language to first order logic. I could not find works that were tested on Claude and more recent versions of the GPTx, Claude and other LLMs, have been upgraded. They are known to generate nearly accurate First Order Logic forms, clausal forms. The explicit-reasoning tasks still require hand-crafted domain-specific rules and logic that are difficult to scale and generalize to other applications [23]. This also brings out the problem of lack of common consensus for measuring reasoning capabilities in LLMs [23].

First Order Logic as a language to represent Natural Language:

Natural Languages are known to have inherent expressive power [1]. First Order Logic provides a mechanism for systems to better understand unstructured text towards language understanding tasks. [3] Capturing the expressive power of natural language by using First Order logic representations is another commonly followed direction [1] [2].

Time and Space complexity, State Space

W.r.t to time, only one pair at a time is resolved, when there are multiple complementary literals and two statements can get illegal resolution depending on the sequence of resolution. The order matters. When we do hand-solve, we have some clues to make it more general, but it is difficult to stop the prover as the nested functions for a single predicate can grow. The worst case size can grow exponentially and the prover can run forever. Size of history increases as the resolution explores all possible combinations. Space depends on the number of clauses and complexity of the clauses (depth of nested clause). We did not consider equality and it seems that it reduces the number of possibilities or paramodulation possibilities for this specific use case of considering equality.

Resolution is refutation complete, if a set of clauses, that are in first order clausal normal forms, are unsatisfiable, then the resolution derives empty clauses and verifies unsatisfiability. But Resolution is NOT refutation complete if the set of clauses may be satisfiable, that is there is a clause that is a logical consequence of the set of clauses, as it may never terminate. If C is a first order clausal normal form that is unsatisfiable, then $C \vdash \perp$, since resolution combined with skolemization and classification is a refutation

complete calculus for First Order Logic.[11] By Lifting lemma and Herbrand's theorem, proof for First-order resolution's refutation completeness can be proved. If a formula is satisfiable, proof search can terminate or go one forever, as there are infinitely many clauses that can be generated by resolution. Therefore resolution is not a decision procedure for First order logic. There may be other forms of resolution [12] but we only use conventional skolemization decomposition for this project. The conventional skolemization decomposition (CSD) are not meaning-preserving clausal forms. [13] LLMs on the other hand, terminate and don't get stuck whether to conclude true, false or inconclusive, despite various other issues surrounding accuracy and planning during deductive reasoning. [7][15]

How to generate a really long knowledge base and convert to Skolemized clausal forms?

1) Creating larger knowledge base and converting to clausal forms - Use LogicLLAMA and Claude AI to create a really large Knowledge base in clausal forms. And evaluating if the resulting clausal forms are correct may be challenging due to hallucinations and other known issues.[7]
2) Future work is focussed on generating dataset of knowledge bases with word sense disambiguation using LLMs and evaluate LLMs.

Example Knowledge Bases, Clausal Forms and proofs

VS

Example Knowledge base 1:

Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bat flies with wings.
Wings are not barrels.
Barrels are not wings.
Bats can fly.
There is a brown bat.

--- query ---
Brown Bat can fly.
--- negated query ---
Brown Bat cannot fly.

Will derive bottom since the negated query is False and the query is True.
There is at least one bat that is brown and bats can fly.

CNF grammar, same as given in :

clause → literal | clause
| literal
literal → predicate
| - predicate
predicate → capitalized-name (term-list)
term-list → term , term-list
| term
term → capitalized-constant-name
| capitalized-function-name (term-list)
| lowercase-variable-name

Skolemized Clausal forms (manual):

1. Mammal(Bat) | Brown(Bat)
2. UsedIn(Bat, Baseball)
3. (∀x (Bat(x) -> Has(x, Barrel))
4. (∀x (Bat(x) -> ∃y (Wing(y) ∧ FliesWith(x,y))))
5. (∀x (Wing(x) -> ¬ Barrel(x))
6. (∀x (Barrel(x) -> ¬ Wing(x))
7. (∀x (Bat(x) -> CanFly(x))
8. (∃x (Bat(x) ∧ Brown(x)))
   --- query ---
   (∃x (Bat(x) ∧ Brown(x) -> CanFly(x)))
   --- negated query ---
   ¬ (∃x (Bat(x) ∧ Brown(x) -> CanFly(x)))

query:

(∃x (Bat(x) ∧ Brown(x) -> CanFly(x))) => ∃x (¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x)))

∃x (¬ Bat(x) ∨ ¬ Brown(x)) ∨ CanFly(x)))

Skolemization constant c ¬ Bat(c) ∨ ¬ Brown(c) ∨ CanFly(c)

Final Clausal form **{** ¬ Bat(c) , ¬ Brown(c)), CanFly(c))) **}**

Negated query:

(∃x (Bat(x) ∧ Brown(x) -> CanFly(x))) => ¬ ∃x (¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x)))

Apply Quantifier Negation ∀x ¬ (¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x)))

∀x (¬¬ (Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x)))

∀x ((Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x)))

((Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x)))

Final Clausal form **{** {Bat(x) }, { Brown(x) }, {¬CanFly(x) } **}**

Proof:

1. Mammal(Bat) | Brown(Bat)
2. UsedIn(Bat, Baseball)
3. ¬Bat(x) ∨ Has(x, Barrel)
4. ¬Bat(x) ∨ Wing(f(x))
5. ¬Bat(x) ∨ FliesWith(x,f(x))
6. ¬Wing(x) ∨ ¬Barrel(x)
7. ¬Barrel(x) ∨ ¬Wing(x)
8. ¬Bat(x) ∨ CanFly(x)
9. ¬Bat(x) ∨ Brown(x)
10. Bat(c)
11. Brown(c)
12. ¬CanFly(c)

Resolve 8 & 10 => ¬Bat(x) ∨ CanFly(x) and Substitute x = c

¬Bat(c) ∨ CanFly(c) Bat(c)

—----------

CanFly(c)

13. CanFly(c)

Resolve 12 & 13 => CanFly(c) ¬CanFly(c)

—----------

[ ]

Therefore it entails that Brown Bat can fly.
Proof steps 4. Not meaning preserving.

Example Output for this problem is:

1: ¬Bat(c) ∨ CanFly(c)
2: Bat(c)
3: CanFly(c)
4: ¬CanFly(c)
1 & 2 give 3: CanFly(c)
3 & 4 give 5: <empty>

Knowledge base 2 (with noise):

Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bats have wings.
Bats can fly.
Flying requires wings.
Wings are not barrels.
Barrels are not wings.
Bat can be elegant.
There is a brown bat.
--- query ---
Elegant Brown Bat has a barrel and can fly.
--- negated query ---
Elegant Brown Bat does not have a barrel or cannot fly.

Will not derive bottom since the negated query is true and the query is false.
There is atleast one bat that is brown, bats can fly, flying requires wings and wings are not barrels and barrels are not wings.

Skolemized Clausal forms (manual):

1. Mammal(Bat)
2. UsedIn(Bat, Baseball)
3. ¬Bat(x) ∨ Barrel(b(x))
4. ¬Bat(x) ∨ Has(x,b(x))
5. ¬Bat(x) ∨ Wing(w(x))
6. ¬Bat(x) ∨ Has(x,w(x))
7. ¬Bat(x) ∨ CanFly(x)
8. ¬CanFly(x) ∨ Wing(wf(x))
9. ¬CanFly(x) ∨ Has(x,wf(x))
10. ¬Wing(x) ∨ ¬Barrel(x)
11. ¬Barrel(x) ∨ ¬Wing(x)
12. Bat(e)
13. Elegant(e)
14. Bat(c)
15. Brown(c)
16. Bat(d)
17. Brown(d)
18. ¬CanFly(d)

Resolve 7 and 16
¬Bat(x) ∨ CanFly(x) and Bat(d)
Substitute x = d

¬Bat(d) ∨ CanFly(d) Bat(d)
—----------
CanFly(d)

19. CanFly(d)

Resolve 19 and 18,

¬CanFly(d) CanFly(d)
—----------
[ ]

Example Output for this problem is:

1: ¬Bat(x) ∨ CanFly(x)
2: Bat(d)
3: ¬CanFly(c)
4: CanFly(c)
1 & 2 give 3: ¬CanFly(c)
3 & 4 give 5: <empty>

True.

What changed from previous version V1:

Updated as per review comments. The section "Why do we want to test the reasoning for inconclusiveness for Explicit reasoning tasks?" and "Intuition and Motivation" sections address the majority of questions about why, intuition, and motivation parts.
The goal, problem statement, input, output, method, results, purpose and choice of knowledge base, rationale about the prompts, methods and details are included. The discussion about Resolution Refutation is also included. I included a STRIPS planning paper that the Professor shared.

- 0.5 please revise + resubmit! I was not able to understand much of this proposal :(
- intuition + motivation behind problem?
- proof of what? what is query?
- how different from asst 6+7
- why do we care? will same proofs not be correct?
- from the input? how?
- how? citation + explanation!
- how?
- difference from 5?
- do you measure this or is it part of LLM output?
- where is bibliography? how does this relate to your project?
- I did not understand the purpose yet - up to here this seems very much like asst 6+7
-

What changed from version V0 (pre proposal):

Removed a lot of stuff. It simplified and removed abstract stuff following guidelines of the professor. I may or may not complete all Large Language Models relevant work

depending on time and especially computational resources as well as the fact that I was able to help getting OpenAI EDU licenses for the classes during last semester and I do not have a license for OpenAI. However I planned my work around other LLMs and I am fine.

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CS 830 Introduction to Artificial Intelligence

Prof. Wheeler Ruml

Project proposal, March 11th 2025, Student: Sushma Anand Akoju

Project design idea

I work at the intersection of natural language understanding and neurosymbolic methods for ethical, responsible and fair use of AI. I was hoping to design a knowledge base similar to the ones designed for assignments 6 and 7. I have been studying about measuring logical entailment over Language models. Since I have been working on first order logic and proofs for sometime now and more recently CS 830 has been working on resolution refutation proofs, **following are the concrete steps I had on mind:**

1) To design knowledge base in natural language (as it was asked in the assignment 6 and 7),
2) To convert knowledge base to First Order Logic (FOL) representation (using an existing work that I implemented a while ago),
3) To convert FOL to clausal form, and
4) Run unify and
5) Run resolution refutation
6) Extract proof steps from the path
7) Use the 2), 3) and 6) to run it through **LogicLLaMa, GPT, Claude AI**
8) Direct Prompt structures to use: **Chain-of-thought & Graph-chain-of-thought**
9) Compare the results about why, how and when do we say the algorithms and language model did well

what does this mean?

what will you measure?

not clear to me

The reason steps 1 through 7 are simple is because we already did the majority of the work for assignments 6 and 7, step 6 is not difficult to capture successful steps and failure cases. Step 7 is a simple prompt design which is already available. It should not take more than 5-30 minutes to record the results from language models.

The most challenging part would be step 8) - the part about reading and writing

a) the why, how and when do we conclude the algorithm did well (by appealing to time complexity, state space, search space and backtracking) versus .
b) Why, how and when do we conclude the language model did well and also attempt to score the performance.

What parts of step 8 are challenging?

1) Not challenging part - Step a) is something we do during the homework assignments and lectures discuss elaborately on this.
2) **Challenging part:** Step b) can be inferred from a few recent research works and I could summarize that to re-run to extract some additional details.

3) **Challenging part:** How to select and organize results from steps a) and b) for comparison.
4) Compare where and how step a) and step b) results would differ and why due to design, complexity, results would be the part requiring more focussed and cautious approach.
5) Cite the references of research works relevant for this.

needed for proposal

What if this project is too small and I would end up completing it earlier?

It is not a small project. But if that happens, I would consider running a moral reasoning example using Cp-nets shared by the professor. By using CP-nets approach of preference (example: https://arxiv.org/pdf/1107.0023 page 141) based conditional modeling of the knowledge graph, I would like to design the trolley problem knowledge base, following all of the aforementioned 8 steps. I am not familiar about which algorithms can be used for equivalent proofs and satisfiability, but I would consider designing the trolley problem knowledge base anyway to run through the same algorithms and **discuss why it would not work. That would allow me to learn about an unknown concept and why aforementioned algorithms would not work for trolley problems designed using CP-nets. This way I would not be overcommitting to learning CP-nets and implementing it but I would be learning why aforementioned approaches may not work for CP-nets type of data format.**